Week 9: Modeling Volatility (ARCH/GARCH)

Kurtosis

An ARCH(1) specification

$$\varepsilon_t = \sigma_t u_t$$

$$u_t \overset{iid}{\sim} N(0,1)$$

$$\boxed{\varepsilon_t | F_{t-1} \sim N(0, \sigma_t^2)}$$

The Box.

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2.$$

Recall (**Week 8**) unconditional moments of the ARCH(1) model

(**5**.) It can be shown that the kurtosis for ε_t is

$$\kappa = 3 \times \frac{(1-\alpha_1^2)}{(1-3\alpha_1^2)} > 3.$$

Thus ε_t is Not Normal

good ⇒ w2/w3 says.

when $0 < \alpha_1 < \sqrt{1/3}$ (proof not supplied).

Moments of the ARCH(1) model

- (**5.**) shows that the unconditional distribution of the $\{\varepsilon_t\}$ from an ARCH model **cannot be a normal distribution... too much kurtosis**.

- Why is this important?

  1. Returns are not normal.
  2. Tails are too thick; i.e., too much kurtosis.
  3. However, we can use a conditional normal random variable to model returns.
  4. So long as we augment this random variable with ARCH-like characteristics.

Moments of the ARCH(1) model

This means that this simple model allows us to capture two important stylized facts of returns data:

(1) Returns have thick tails, i.e., are not (unconditionally) normal;

(2) Risk (or volatility) is time-varying and autoregressive, i.e., returns display ARCH-like behavior;

Estimation of an ARCH model

- MLE chooses c, ϕ_1, α_0 and α_1 to maximise the (log of the) joint probability of the observed data points in our sample (called the log likelihood (LL)).

- For an ARCH(1) model (with AR(1) mean and normal u_t the LL is given by

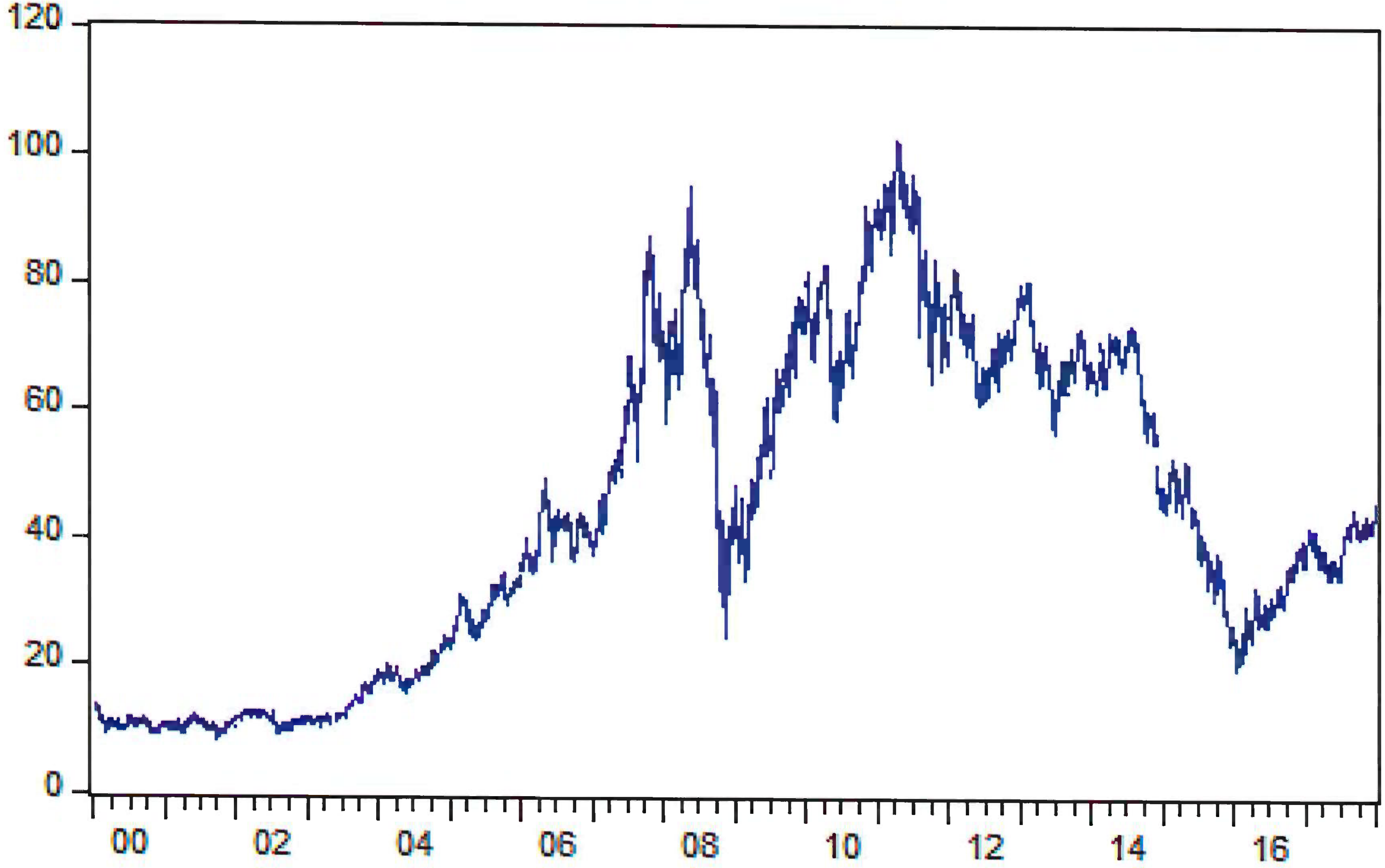
$$-\frac{T}{2}\ln(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\ln\sigma_t^2 - \frac{1}{2}\sum_{t=1}^{T}\frac{(r_t - c - \phi_1 r_{t-1})^2}{\sigma_t^2}, \quad \overset{\max}{= LL}$$

with

$$\sigma_t^2 = \alpha_0 + \alpha_1(r_{t-1} - c - \phi_1 r_{t-2})^2.$$

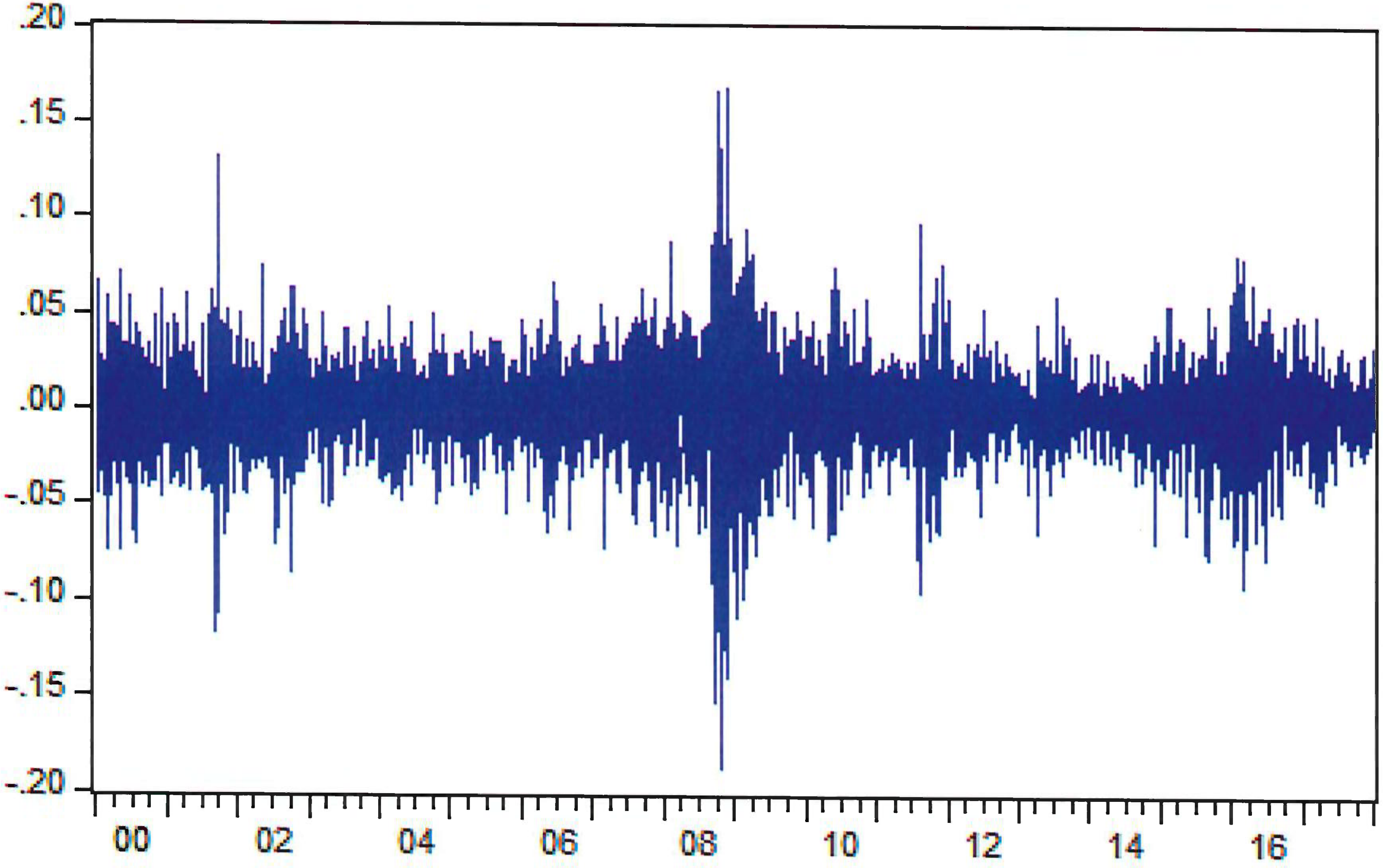
- The form of this LL comes from the AR(1)-ARCH(1) specification and the assumption that $u_t \overset{iid}{\sim} N(0,1)$.

ibm.xts is daily 3 Jan 2000 to 3 Jan 2018



```
# log return
lret <- na.omit(diff(log(ibm.xts)))
colnames(lret) <- "log return"
```

lret = log return



```
# AR(1,0)-ARCH(1)

library(rugarch)
```

xts

name it

Step 1

```
spec_arch1 <- ugarchspec(
  variance.model = list(
    model = "sGARCH",               # standard GARCH
    garchOrder = c(1, 0)            # c(ARCH,GARCH)
  ),
```

ARCH(1)

```
  mean.model = list(
    armaOrder = c(1, 0),            # AR(1)
    include.mean = TRUE
  ),
```

AR(1)

```
  distribution.model = "norm"      # normal errors
)
```

function to estimate

Step 2

```
# Estimate the AR-GARCH models
fit_arch1 <- ugarchfit(
  spec = spec_arch1,
  data = lret
)
```

data.

$\hat{E}(Y_t) = \hat{\mu} = \dfrac{c}{1-\phi_1}$

Slide 12 .

```
> fit_arch1@fit$matcoef
```

not intercept	Estimate	Std. Error	t value	Pr(>|t|)
mu ($\hat{\mu}$)	0.0006500238	3.252055e-04	1.998810	0.04562896
ar1 ($\hat{\phi}_1$)	0.0199421101	1.669515e-02	1.194485	0.23228834
omega ($\hat{\alpha}_0$)	0.0004107126	1.179327e-05	34.826010	0.00000000
alpha1 ($\hat{\alpha}_1$)	0.3042729939	2.621725e-02	11.605833	0.00000000

$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2$

$\downarrow$

0.9 not 0.3

Environment | History | Connections | Tutorial

Import Dataset ▾ | 342 MiB ▾

R ▾ | Global Environment ▾

```
fit_arch1   name it          Formal class  uGARCHfit
  ..@ fit  :List of 27
  .. ..$ hessian        : num [1:4, 1:4] 9472429 647 2623065 -4300 647 ...
  .. ..$ cvar           : num [1:4, 1:4] 1.06e-07 -4.37e-09 -1.05e-10 3.52e-0
  .. ..$ var            : num [1:4529] 0.000594 0.000782 0.000443 0.0005 0.00
1 .. ..$ sigma   σ̂_t    : num [1:4529] 0.0244 0.028 0.021 0.0224 0.0414 ...
                                      σ̂_1    σ̂_2
  .. ..$ condH          : num 6.81
  .. ..$ z              : num [1:4529] 1.434 0.366 -0.814 2.925 -0.26 ...
2 .. ..$ LLH            : num 10639
  .. ..$ log.likelihoods: num [1:4529] -1.77 -2.59 -2.61 1.4 -2.23 ...
3 .. ..$ residuals      : num [1:4529] 0.0349 0.0102 -0.0171 0.0654 -0.0108 .
  .. ..$ coef           : Named num [1:4] 0.00065 0.019942 0.000411 0.304273
  .. .. ..- attr(*, "names")= chr [1:4] "mu" "ar1" "omega" "alpha1"
  .. ..$ robust.cvar    : num [1:4, 1:4] 1.28e-07 4.20e-06 -2.32e-09 8.99e-06
  .. ..$ A              : num [1:4, 1:4] 2091.506 0.143 579.171 -0.949 0.143
  .. ..$ B              : num [1:4, 1:4] 2385.5 23.4 -8312.5 16.8 23.4 ...
  .. ..$ scores         : num [1:4529, 1:4] -59.45 -24.6 35.6 -205.36 -4.46 .
  .. .. ..- attr(*, "dimnames")=List of 2
  .. .. .. ..$ : NULL
  .. .. .. ..$ : chr [1:4] "mu" "ar1" "omega" "alpha1"
```

continued

```
 .. ..$ A              : num [1:4, 1:4] 2091.506 0.143 579.171 -0.949 0.14
 .. ..$ B              : num [1:4, 1:4] 2385.5 23.4 -8312.5 16.8 23.4 ...
 .. ..$ scores         : num [1:4529, 1:4] -59.45 -24.6 35.6 -205.36 -4.46
 .. .. ..- attr(*, "dimnames")=List of 2
 .. .. .. ..$ : NULL
 .. .. .. ..$ : chr [1:4] "mu" "ar1" "omega" "alpha1"
 .. ..$ se.coef        : num [1:4] 3.25e-04 1.67e-02 1.18e-05 2.62e-02
 .. ..$ tval           : Named num [1:4] 2 1.19 34.83 11.61
 .. .. ..- attr(*, "names")= chr [1:4] "mu" "ar1" "omega" "alpha1"
3.. ..$ matcoef        : num [1:4, 1:4] 0.00065 0.019942 0.000411 0.304273
 .. .. ..- attr(*, "dimnames")=List of 2
 .. .. .. ..$ : chr [1:4] "mu" "ar1" "omega" "alpha1"
 .. .. .. ..$ : chr [1:4] " Estimate" " Std. Error" " t value" "Pr(>|t|)"
```

slide 11   name it.

```
> # residuals
> res_arch1 <- residuals(fit_arch1)
> colnames(res_arch1) <- "ehat ARCH"
> tail(res_arch1)
              ehat ARCH
2017-12-26  0.007137154
2017-12-27  0.008427157
2017-12-28  0.012418597
2017-12-29 -0.009345413
2018-01-02  0.032257987
2018-01-03  0.002700624
```

4529   slide 19

$\hat{e}_{4529}$ (last obs).

name

```
> sigma2hat_arch1 <- sigma(fit_arch1)^2
> tail(sigma2hat_arch1)
                   [,1]
2017-12-26 0.0004124374
2017-12-27 0.0004262119
2017-12-28 0.0004323211
2017-12-29 0.0004576380
2018-01-02 0.0004372868
2018-01-03 0.0007273323
```

→ $\hat{\sigma}^2_{4529}$

last obs.

```
> tail(ibm.xts)
             IBM
2017-12-26 45.35
2017-12-27 45.77
2017-12-28 46.38
2017-12-29 45.99
2018-01-02 47.52
2018-01-03 47.71
```

$ (handwritten beside IBM)

P_{4529}

01-04 Forecast next day.

```
> tail(lret)
             log return
2017-12-26  0.007747643
2017-12-27  0.009218722
2017-12-28  0.013239499
2017-12-29 -0.008444328
2018-01-02  0.032726650
2018-01-03  0.003990324
```

← r_{4529}

01-04 Forecast.

↓

4 Jan

Mean-Adjusted AR models

The AR(1) model is written in mean-adjusted form as

$$y_t - \hat{\mu} = \hat{\phi}(y_{t-1} - \hat{\mu}) + \hat{e}_t,$$

where

$$\hat{\mu} = 0.0006500238$$ slide 10

and

$$\hat{\phi} = 0.0199421101.$$

Mean-Adjusted AR Models

To rewrite the model in the usual intercept form,

$$y_t = c + \phi y_{t-1} + e_t,$$

we expand the mean-adjusted AR(1) model as follows:

R

$$y_t = \mu + \phi y_{t-1} - \phi\mu + e_t. \quad \leftarrow \quad Y_t - \mu = \phi(Y_{t-1} - \mu) + e_t$$

Therefore,

$$y_t = \mu(1 - \phi) + \phi y_{t-1} + e_t.$$

Hence,

$$c = \mu(1 - \phi).$$

$$\mu = \frac{c}{1-\phi} \Rightarrow E(Y_t) = LR.$$

Mean-Adjusted AR Models

Using the estimated values,

$$\hat{c} = 0.0006500238(1 - 0.0199421101), \quad \Rightarrow c = \hat{u}(1-\hat{\phi})$$

so

$$\hat{c} \approx 0.000637061.$$

intercept

Thus, the AR(1) model in the usual intercept form is

$$y_t = \underset{\hat{c}}{0.00064} \underset{+}{+} \underset{\hat{\phi}_1}{0.0199421101}\underset{y_{t-1}}{y_{t-1}} + \hat{e}_t.$$

The unconditional mean is

$$\hat{u} = \frac{\hat{c}}{1-\hat{\phi}} = \frac{0.000637061}{1 - 0.0199421101} = 0.0006500238.$$

Forecasting: an AR(1)-ARCH(1) model

Fitted model for IBM returns:

$$r_t = \underbrace{0.00064}_{\hat{c}} + \underbrace{0.01994}_{\hat{\phi}} r_{t-1} + \hat{\varepsilon}_t,$$

with

omega alpha1 slide 10

$$\hat{\varepsilon}_t = \hat{\sigma}_t \hat{u}_t, \quad \underset{T+1}{\hat{\sigma}_t^2} = 0.00041 + 0.30427 \underset{T}{\hat{\varepsilon}_{t-1}^2}.$$

LL (Log likelihood) = 10638.59

Daily IBM log returns: 4 Jan 2000 to 3 Jan 2018 (4529 obs)

Let's forecast 4 Jan 2018

$\mathcal{F}_{4529}$: We have:

$r_T =$ $r_{4529} = 0.004$, $\widehat{\varepsilon}_{4529} = 0.0027$, $\widehat{\sigma}^2_{4529} = 0.00073$, and $P_{4529} = \$47.71$.

slide 15/14

Forecasting with an AR(1)-ARCH(1) model

Forecasts:

Conditional mean:

$$\hat{\mu}_{T+1|T} = \widehat{E}[\underbrace{r_{4530}}_{T+1}|\underbrace{\mathcal{F}_{4529}}_{T}] = 0.00064 + 0.01994 \times (\overbrace{0.004}^{r_T})$$

$$= 0.0007$$

⇒ 1-step ahead forecast.

Conditional variance:

$$\hat{\sigma}^2_{T+1|T} = \hat{\mathbb{V}}[r_{4530}|\mathcal{F}_{4529}] = \hat{\mathbb{V}}[\varepsilon_{4530}|\mathcal{F}_{4529}]$$

$$= \overset{\hat{\alpha}_0}{0.00041} + \overset{\hat{\alpha}_1}{0.30427} \times (0.0027)^2$$

$$= 0.00041$$

→ 1-step forecast

$\hat{\varepsilon}_T = \hat{\varepsilon}_{4529}$

⇒ Volatility:

$$\widehat{\sigma}_{4530}[r_{4530}|F_{4529}] = \sqrt{0.0004} = 0.02 = \hat{\sigma}_{T+1|T}$$

Forecasting using R

```
# h-step ahead forecast                    name
h <- 1
fc_arch1 <- ugarchforecast(fit_arch1, n.ahead = h)

# Mean forecasts
mu_fc_arch1 <- as.numeric(fitted(fc_arch1))
mu_fc_arch1 = 0.0007

# "Volatility" forecasts
sigma_fc_arch1 <- as.numeric(sigma(fc_arch1))
sigma_fc_arch1
# Variance forecasts
var_fc_arch1 <- sigma_fc_arch1^2
var_fc_arch1 = 0.00041
```

Forecasting with an AR(1)-ARCH(1) model

If we assume that $u_t \sim N(0,1)$ so that $\varepsilon_t | \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$ then the conditional distribution of returns will also be normally distributed.

We can compute the 95% prediction interval for the one-step-ahead forecast of r_t.

A 95% prediction interval for r_{4530} would be

$$\underbrace{0.0007166364}_{\hat{\mu}_{T+1|T}} \pm 1.96 \times \underbrace{0.02032072}_{\hat{\sigma}_{T+1|T}}$$

giving the interval

$$[-0.0391, 0.0405] \Rightarrow$$

desire a very narrow interval.

interval : $\hat{\mu}_{T+1|T} \pm 1.96\, \hat{\sigma}_{T+1|T}$

Forecasting with an AR(1)-ARCH(1) model

If we assume that $u_t \sim N(0, 1)$ so that $\varepsilon_t | \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$ then the conditional distribution of returns will also be normally distributed.

Recall $P_{4529} = \$47.71$ today

We can form prediction intervals for the **next price**

A 95% prediction interval for P_{4530} is

1-step   1-step ⇒ slide 22

$$\left[\underbrace{\$47.71}_{P_T} e^{\overbrace{-0.0391}^{\text{1-step}}},\ \underbrace{\$47.71}_{P_T} e^{\overbrace{0.0405}^{\text{1-step}}}\right] = [\$45.88,\ \$49.68]$$

$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

$e^{r_t} = \frac{P_t}{P_{t-1}} \Rightarrow P_t = P_{t-1} e^{r_t}$

$P_{t+1} = P_t e^{r_{t+1}}$

Diagnostics for an ARCH model

- We can use the ARCH LM test to determine if serial correlation in $\widehat{u}_t^2$ remains.
  - if there appears to be serial correlation remaining in the squared residuals,
  - $\Rightarrow$ we can improve the volatility model by including more ARCH terms.

left-over ARCH effect.

ARCH test

ARCH(1) is wrong then
then $\hat{u}_t$ is wrong too.

$\hat{u}_t$ not $\hat{\varepsilon}_t$

name it

```
> res_s.arch1     <- residuals(fit_arch1, standardize = TRUE)
> FinTS::ArchTest(res_s.arch1, lags = 4)

        ARCH LM-test; Null hypothesis: no ARCH effects

data:  res_s.arch1
Chi-squared = 303.02, df = 4, p-value < 2.2e-16
```

$$\hat{u}_t = \alpha_0 + \alpha_1 \hat{u}^2_{t-1} + \cdots + \alpha_4 \hat{u}^2_{t-4}$$

$H_0: \alpha_1 = \alpha_2 = \cdots = \alpha_4 = 0$

H_A: at least one

LM = 303.02 , p-value = 0

Reject H_0 ⇒ ARCH 1 is mis-specified.

ARCH(q)

1. In many cases we can improve our volatility model by working with an ARCH(q) model.

2. ARCH(q) takes $\varepsilon_t = u_t \sigma_t,\ u_t \sim i.i.d.N(0,1)$ and

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \ldots + \underbrace{\alpha_q \varepsilon_{t-q}^2}.$$

Solution => add more lags.

- In many cases, especially with **higher frequency data**, we need q to be very large;

  1. Otherwise $\widehat{u}_t^2$ will be auto-correlated.
  2. Suggests that we could improve our modeling of conditional variance.
  3. This can be solved using GARCH models (generalized ARCH).

- The $\widehat{u}_t$ do not appear to have come from a normally distributed u_t.

  1. Can be fixed by changing the assumptions about u_t.
  2. E.g., to get more kurtosis, we could assume that the u_t have a t-distribution.
  3. This involves maximising a different likelihood.

$u_t \sim iid\, N(0,1)$

Student-t Distribution

```
> fit_arch1_t@fit$matcoef
            Estimate   Std. Error   t value     Pr(>|t|)
mu      0.0007977183 2.906414e-04  2.744682 0.006056951
ar1    -0.0201519356 1.689249e-02 -1.192952 0.232888013
omega   0.0004235392 1.834514e-05 23.087267 0.000000000
alpha1  0.2881657780 3.291013e-02  8.756143 0.000000000
shape   5.0133763383 3.739183e-01 13.407680 0.000000000
```

d.f.

Student-t Distribution

"normal"

slide-9

→ Student-t ⇒ d.f.

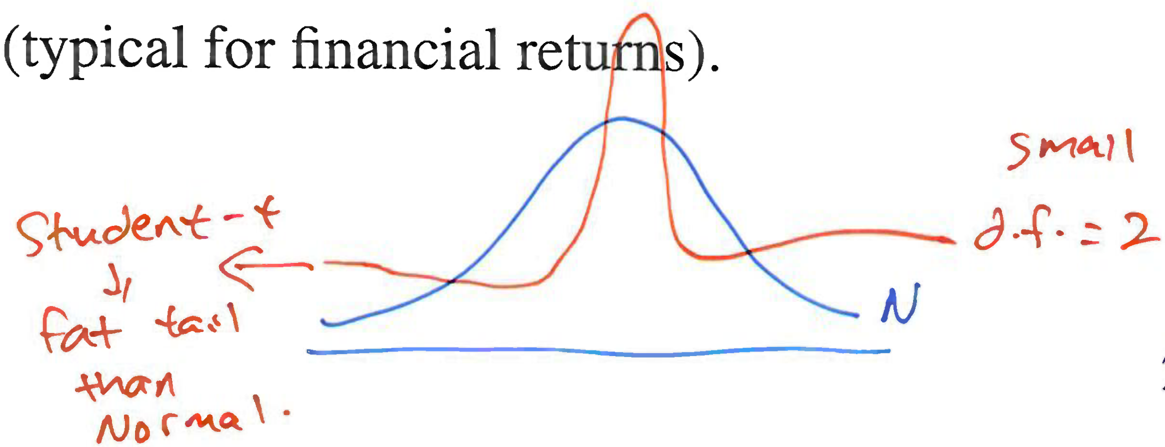
In 'rugarch' with *distribution.model* = "*std*", the parameter *shape* is the degrees of freedom v of the Student-t distribution for the model errors.

In the R output: *shape* $= 5.0134 \approx 5$

What that means:
Smaller $v \rightarrow$ heavier tails $\rightarrow$ more probability of large shocks/outliers than Normal.

As $v \rightarrow \infty$ Student-t approaches the Normal.

So $v \approx 5$ is heavy-tailed (typical for financial returns).



GARCH models

1:03
1:08 back.

ARCH models have difficulty modelling the autocorrelation in the squared residuals.

GARCH models generalise ARCH models to include lags of σ_t^2 in the conditional variance equation as well as lags of ε_t^2, and this often alleviates this difficulty.

A GARCH(p,q) model is given by

$$\sigma_t^2 = \alpha_0 + \underbrace{\alpha_1\varepsilon_{t-1}^2 + \ldots + \alpha_q\varepsilon_{t-q}^2}_{\text{Bob Engle}} + \underbrace{\beta_1\sigma_{t-1}^2 + \ldots + \beta_p\sigma_{t-p}^2}_{\text{GARCH(p)}}.$$

ARCH(q)

Can I do this

σ_{t-1}^2 σ_{t-2}^2

GARCH(1,1) is the Best model.
for a lot of stocks.

GARCH models

GARCH(1,1) $\Rightarrow$ $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$

$\alpha_1 \approx 0.1$ β_1 about 0.9 wow!

A GARCH(p,q) model

$$\sigma_t^2 = \alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \ldots + \alpha_q\varepsilon_{t-q}^2 + \beta_1\sigma_{t-1}^2 + \ldots + \beta_p\sigma_{t-p}^2. \quad \sigma_t^2 > 0$$

1 >0 >0 >0 >0 >0

Estimation in R is not guarantee to be positive.

We assume that
(i) all $\alpha_i > 0$ and all $\beta_j > 0$ to ensure that $\sigma_t^2 > 0$, and

(ii) that $\sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j < 1$ to ensure that $\mathbb{V}(\varepsilon_t) > 0$.

$q = p = 1$ 2 $\alpha_1 + \beta_1 < 1 \Rightarrow V(\varepsilon_t) > 0$ in practice $\alpha_1 + \beta_1$ close to one.

The α_i and β_j determine how past news affect current volatility and this persistence is often measured by $\gamma = \sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j$.

If α_0 is the only non-zero parameter then we have constant volatility.

$$V(\varepsilon_t) = \frac{\alpha_0}{1-\alpha_1-\beta_1} = \frac{\alpha_0}{1-\underbrace{(\alpha_1+\beta_1)}_{<1}} > 0$$

Conditional moments of GARCH(p,q) errors

Recall that $\varepsilon_t = \sigma_t u_t$ with $u_t \sim N(0,1)$, so the standardised error, $\frac{\varepsilon_t}{\sigma_t} \sim N(0,1)$ and the conditional distribution $\varepsilon_t|\mathcal{F}_{t-1} \sim N(0,\sigma_t^2)$.

Thus:

1. $E(\varepsilon_t|\mathcal{F}_{t-1}) = 0$
2. $\mathbb{V}(\varepsilon_t|\mathcal{F}_{t-1}) = \sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2$
3. $Cov(\varepsilon_t, \varepsilon_{t-k}|\mathcal{F}_{t-1}) = 0$ for $k \geq 1$
4. The skewness of $\varepsilon_t|\mathcal{F}_{t-1}$ is zero; and
5. The kurtosis of $\varepsilon_t|\mathcal{F}_{t-1}$ is 3.

All but point 3 follow directly from the fact that $\varepsilon_t|\mathcal{F}_{t-1} \sim N(0,\sigma_t^2)$.

Conditional moments of GARCH(p,q) errors

The proof for point 3 relies on the fact that ε_{t-k} is known at time $t-1$ so

$$Cov(\varepsilon_t, \varepsilon_{t-k}|\mathcal{F}_{t-1}) = E(\varepsilon_t\varepsilon_{t-k}|\mathcal{F}_{t-1}) = \varepsilon_{t-k}E(\varepsilon_t|\mathcal{F}_{t-1}) = 0.$$

Unconditional moments of the GARCH(p, q) errors

(1) $E(\varepsilon_t) = 0$

(2) $\mathbb{V}(\varepsilon_t) = \alpha_0/(1 - \sum_{i=1}^{q} \alpha_i - \sum_{j=1}^{p} \beta_j)$

(3) $Cov(\varepsilon_t, \varepsilon_{t-k}) = 0$ for all $k \geq 1$

(4) $E(\varepsilon_t^3) = 0$

(5) It can be shown that kurtosis $\kappa > 3$ (proof not supplied).

Unconditional moments of the GARCH(p, q) errors

(1) $E(\varepsilon_t) = 0$

Proof.

$E(\varepsilon_t) = E[E(\varepsilon_t|\mathcal{F}_{t-1})]$ by the L.I.E., and since $\varepsilon_t|\mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$ it follows that $E(\varepsilon_t|\mathcal{F}_{t-1}) = 0$.

Therefore $E(\varepsilon_t) = E(0) = 0$.

$\square$

(2) $\mathbb{V}(\varepsilon_t) = \alpha_0/(1 - \sum_{i=1}^{q} \alpha_i - \sum_{j=1}^{p} \beta_j)$

Proof.

We first note that $E(\varepsilon_t) = 0$ implies that $\mathbb{V}(\varepsilon_t) = E(\varepsilon_t^2)$, and then by the L.I.E.

$$E(\varepsilon_t^2) = E[E(\varepsilon_t^2|\mathcal{F}_{t-1})] = E(\sigma_t^2).$$

Next we note that since $\varepsilon_t|\mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$ and $E(\varepsilon_t|\mathcal{F}_{t-1}) = 0$, it follows that $\sigma_t^2 = \mathbb{V}(\varepsilon_t|\mathcal{F}_{t-1}) = E(\varepsilon_t^2|\mathcal{F}_{t-1})$. Hence

$$\begin{aligned}
\mathbb{V}(\varepsilon_t) &= E(\varepsilon_t^2) \\
&= E(\sigma_t^2) \\
&= \alpha_0 + \sum_{i=1}^{q} \alpha_i E(\varepsilon_{t-i}^2) + \sum_{j=1}^{p} \beta_j E(\sigma_{t-j}^2).
\end{aligned}$$

$\square$

Proof.
But $E(\varepsilon_{t-i}^2) = E(\sigma_{t-j}^2) = \mathbb{V}(\varepsilon_t)$ since ε_t is stationary, so

$$\mathbb{V}(\varepsilon_t) = \alpha_0 + \mathbb{V}(\varepsilon_t)\left[\sum_{i=1}^{q}\alpha_i + \sum_{j=1}^{p}\beta_j\right].$$

Solving this, we get

$$\mathbb{V}(\varepsilon_t) = \alpha_0 \Big/ \left(1 - \sum_{i=1}^{q}\alpha_i - \sum_{j=1}^{p}\beta_j\right).$$

□

Unconditional moments of the GARCH(p,q) errors (cont)

(3) $Cov(\varepsilon_t, \varepsilon_{t-k}) = 0$ for all $k \geq 1$

Proof.

Note

$$Cov(\varepsilon_t, \varepsilon_{t-k}) = E[(\varepsilon_t - E(\varepsilon_t))(\varepsilon_{t-k} - E(\varepsilon_{t-k}))] = E[(\varepsilon_t)(\varepsilon_{t-k})],$$

since $E(\varepsilon_t) = E(\varepsilon_{t-k}) = 0$. Further, the L.I.E. implies that

$$E[(\varepsilon_t)(\varepsilon_{t-k})] = E[E\{\varepsilon_t \varepsilon_{t-k}\} | \mathcal{F}_{t-1})].$$

Next we note that $\varepsilon_{t-k} | \mathcal{F}_{t-1}$ is known for $k \geq 1$, so

$$E[E\{\varepsilon_t \varepsilon_{t-k}\} | \mathcal{F}_{t-1})] = E[(\varepsilon_{t-k}) E\{\varepsilon_t | \mathcal{F}_{t-1}\}] = 0,$$

(since $E\{\varepsilon_t | \mathcal{F}_{t-1}\} = 0$.)

$\square$

Unconditional moments of the GARCH(p,q) errors (cont)

(4) $E(\varepsilon_t^3) = 0$

Proof.

We firstly note that, by the L.I.E.,

$$E(\varepsilon_t^3) = E[E(\varepsilon_t^3 | \mathcal{F}_{t-1})].$$

Next under the assumption that $u_t \sim i.i.d.N(0,1)$, we have

$$\varepsilon_t | \mathcal{F}_{t-1} \sim N(0, \sigma_t^2) \text{ and } E(\varepsilon_t^3 | \mathcal{F}_{t-1}) = 0.$$

Then, $E(\varepsilon_t^3) = E[E(\varepsilon_t^3 | \mathcal{F}_{t-1})] = E[0] = 0 \implies$ unconditional distribution is symmetric. $\square$

Unconditional moments of the GARCH(p,q) errors (cont)

(5) It can be shown that kurtosis $\kappa > 3$ (proof not supplied).

- Clearly this should be true.
- If $\beta = 0$, we get back $ARCH(1)$.
- ARCH(1) has $\kappa > 3$.
- Therefore, for $0 \leq \beta \leq 1$, GARCH(1,1) has $\kappa > 3$.

GARCH(p,q) Models

- What conditions are needed to ensure $\sigma_t^2 > 0$?
  1. $\alpha_0 > 0$;
  2. $\alpha_i \geq 0$;
  3. $\beta_j \geq 0$;

- For GARCH(1,1)model:

$$\epsilon_t = \sigma_t u_t \quad \sigma_t^2 = \underset{>0}{\alpha_0} + \underset{\geq 0}{\alpha_1} \epsilon_{t-1}^2 + \underset{\geq 0}{\beta_1} \sigma_{t-1}^2,$$

this amounts to

1. $\alpha_0 > 0$;
2. $\alpha_1 \geq 0$;
3. $\beta_1 \geq 0$;

Estimation of a GARCH model

- OLS does not provide optimal estimators in this setting so we use MLE (maximum likelihood estimation), as for ARCH models.
- How to estimate GARCH models in R?

```r
# ARMA(1,0)-GARCH(1,1)
```

S1

```r
spec_garch11 <- ugarchspec(
  variance.model = list(
    model = "sGARCH",                    # standard GARCH
    garchOrder = c(1, 1)                 # c(ARCH,GARCH)
  ),
```

c(q,p)

```r
  mean.model = list(
    armaOrder = c(1, 0),                 # AR(1)
    include.mean = TRUE
  ),
  distribution.model = "norm"
)
```

S2

```r
# Estimate the ARMA-GARCH models
fit_garch11 <- ugarchfit(
  spec = spec_garch11,
  data = lret
)
```

data

```
> fit_garch11@fit$matcoef
             Estimate   Std. Error     t value   Pr(>|t|)
mu       5.420103e-04 2.746391e-04    1.973537 0.04843443
ar1     -2.770662e-02 1.553571e-02   -1.783414 0.07451887
omega    5.565428e-06 2.361384e-06    2.356851 0.01843066
alpha1   6.528361e-02 7.364966e-03    8.864074 0.00000000
beta1    9.243447e-01 9.173868e-03  100.758445 0.00000000
```

$\hat{\beta}_1$

0.924 wow!

$$\sigma_t^2 = \hat{\alpha}_0 + \alpha_1 \boxed{} + \beta_1 \underbrace{\sigma_{t-1}^2}$$

$\boxed{\text{want } \hat{\sigma}_t^2 \Rightarrow \text{a number}}$

don't care.

Estimated (Fitted) Conditional Variance

$\hat{\sigma}_t^2$

```
sigma2hat_garch11 <- sigma(fit_garch11)^2
tail(sigma2hat_garch11)

plot(sigma2hat_garch11)
```
next slide

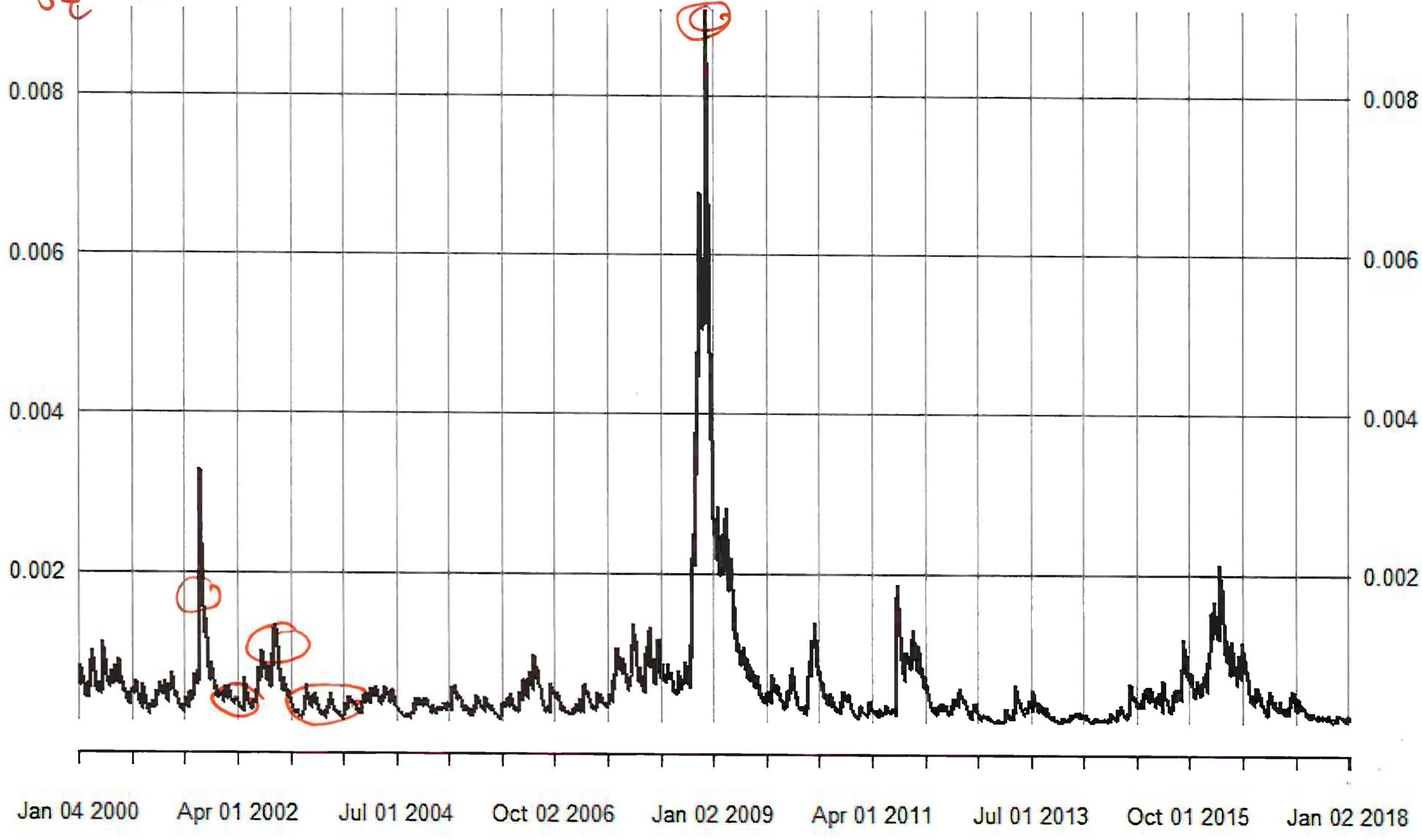
```
plot(sigma(fit_garch11)*100)
```
slide 47.

$\hat{\sigma}_t$

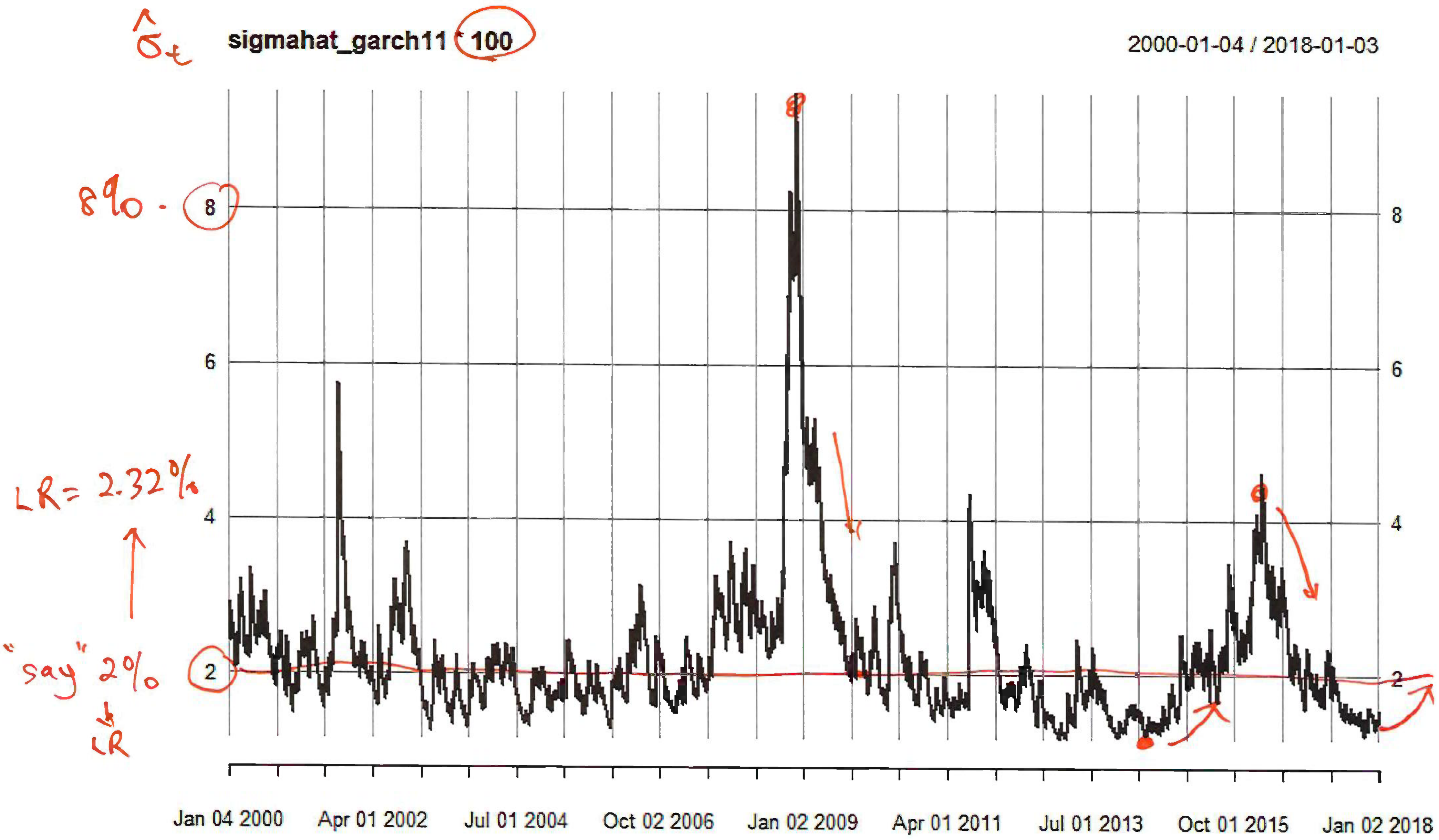
Estimated (fitted) Conditional Variance

$(\%)^2$ $\hat{\sigma}_t^2$

0.008



Estimated (Fitted) Conditional Volatility



Comments on our GARCH(1,1) model

$\widehat{\alpha}_1 \ll \widehat{\beta}_1$. This is typical of GARCH(1,1) models.

Persistence $\widehat{\gamma} = \widehat{\alpha}_1 + \widehat{\beta}_1 = 0.06528 + 0.92434 = 0.98962$ < 1

LR $\Rightarrow \widehat{\mathbb{V}}(\varepsilon_t) = \dfrac{\widehat{\alpha}_0}{1 - \widehat{\alpha}_1 - \widehat{\beta}_1} = \dfrac{5.565 \times 10^{-6}}{1 - 0.98962} = \dfrac{0.000005565}{0.01038} = 0.000537.$

Long run variance

The implied long run variance is 0.000537 and thus the implied long run volatility is

$$\sqrt{\hat{V}(\varepsilon_t)} = \sqrt{\frac{5.565 \times 10^{-6}}{1 - 0.98962}} = \sqrt{0.000537} = 0.02317.$$

So the unconditional volatility is about 2.32%.

Long-Run Volatility

Although financial markets may experience excessive volatility every now and again, volatility appears to eventually settle down to a long run level. The long run level is the unconditional variance.

For the GARCH(1,1) model the long-run variance is

$$\mathbb{V}(\varepsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}.$$

In this case, the volatility is always pulled toward this long run level; that is volatility 'mean reverts' to its long-run level.

```
> # ehat
> res_garch11 <- residuals(fit_garch11)
> colnames(res_garch11) <- "ehat GARCH"
> tail(res_garch11)
             ehat GARCH
2017-12-26  0.007153697
2017-12-27  0.008876356
2017-12-28  0.012937891
2017-12-29 -0.008634534
2018-01-02  0.031935659
2018-01-03  0.004340041
```

$= \hat{\varepsilon}_{4529}$ slide 53

```
> # variance sigma^2
> sigma2hat.garch11  <- sigma(fit_garch11)^2
> colnames(sigma2hat.garch11) <- "sigma2hat GARCH"
> tail(sigma2hat.garch11)
           sigma2hat GARCH
2017-12-26    0.0002023347
2017-12-27    0.0001959333
2017-12-28    0.0001918190
2017-12-29    0.0001938001
2018-01-02    0.0001895707
2018-01-03    0.0002473760
```

$= \hat{\sigma}^2_{4529}$ slide 53

Forecasting with an AR(1)-GARCH(1,1) model

not mu in R

Estimated model:

$$r_t = \overset{\hat{c}}{0.000557} - \overset{ar1}{0.0277} r_{t-1} + \hat{\varepsilon}_t, \quad \hat{\varepsilon}_t = \hat{\sigma}_t \hat{u}_t,$$

$$\hat{\sigma}_t^2 = \underbrace{.0000056}_{\text{omega}} + \underset{\text{alpha1}}{0.06528}\hat{\varepsilon}_{t-1}^2 + \underset{\text{beta1}}{0.92434}\hat{\sigma}_{t-1}^2.$$ slide 44.

LL (Log likelihood) = 11045.76

Daily IBM log returns: 4 Jan 2000 to 3 Jan 2018 (4529 obs)

$$\mathcal{F}_{4529}: \underset{r_T}{r_{4529} = 0.004}, \ \underbrace{\hat{\varepsilon}_{4529} = 0.00434}_{\hat{\varepsilon}_T}, \ \underbrace{\hat{\sigma}_{4529}^2 = 0.000247}_{\hat{\sigma}_T^2}$$ garch(1,1)

Forecasting with an AR(1)-GARCH(1,1) model

- One-step ahead forecast of r_t:

- $\hat{\mu}_{T+1|T}$ $E[r_{4530}|\mathcal{F}_{4529}] = 0.000557 - 0.0277 \times (\overset{r_T}{0.004}) = 0.0004$ = 1-step.

- $\hat{\sigma}^2_{T+1|T}$ $\mathbb{V}[r_{4530}|\mathcal{F}_{4529}] = \mathbb{V}[\varepsilon_{4530}|\mathcal{F}_{4529}] = 0.0000056 + 0.06528 \times (\underbrace{0.00434}_{\hat{\varepsilon}_T^2})^2 + 0.92434 \times (\underbrace{0.000247}_{\hat{\sigma}_T^2}) = 0.000235$

- so $\sqrt{\mathbb{V}[r_{4530}|\mathcal{F}_{4529}]} = \sqrt{0.000235} = 0.0153$

Forecasting using R

```
> h <- 1
> fc_garch11 <- ugarchforecast(fit_garch11, n.ahead = h)
>
> # Mean forecasts
> mu_fc_garch11 <- as.numeric(fitted(fc_garch11))
> mu_fc_garch11
[1] 0.0004464693
>
> # Volatility forecasts
> sigma_fc_garch11 <- as.numeric(sigma(fc_garch11))
> sigma_fc_garch11
[1] 0.01534457
> # Variance forecasts
> var_fc_garch11 <- sigma_fc_garch11^2
> var_fc_garch11
[1] 0.0002354558
```

name it.

$\rightarrow \hat{\mu}_{T+1|T}$ Slide 54

$\Rightarrow \hat{\sigma}^2_{T+1|T}$ Slide 54.

Our model $\Rightarrow E[r_{4530}|\mathcal{F}_{4529}] = 0.0004$ & $\sigma[r_{4530}|\mathcal{F}_{4529}] = 0.0153$

If we then assume that $u_t \sim N(0,1)$ so that $\varepsilon_t|\mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$ then the conditional distribution of returns will also be normally distributed.

Compute the 95% prediction interval for the one-step-ahead forecast of r_t as

$$0.0004464693 \pm 1.96 \times 0.01534457 \implies [-0.0296, 0.0305]$$

$\hat{\mu}_{T+1|T} \pm 1.96 \; \hat{\sigma}_{T+1|T}$

ARCH vs GARCH prediction intervals

One-step-ahead prediction interval for ARCH(1) is

$$[-0.03911125, 0.04054452]$$ slide 22

One-step-ahead prediction interval for GARCH(1,1) is

$$[-0.0296, 0.0305]$$

gives a more percise forecast .

So GARCH(1,1) has the narrower one-step ahead prediction interval than ARCH(1).

Comparing ARCH/GARCH Models

GARCH(1,1) $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$

ARCH(1) $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$

- Likelihood Ratio test.

- ARCH(q) models are nested in GARCH(p,q) models: when $\beta_1 = 0$ GARCH(1,1) model nests (becomes) ARCH(1) model.
  1. $\beta_1 = \cdots = \beta_p = 0 \implies ARCH(q)$.
  2. Can jointly test for significance of β_js

- If the ARCH specification in question is nested in the GARCH model, can use:

$H_0: \beta_1 = 0$

$$LR = 2 * \left(\text{LL}(\hat{\theta}_G) - \text{LL}(\hat{\theta}_A)\right) > 0$$

  1. $\text{LL}(\hat{\theta}_G)$: likelihood from GARCH model
  2. $\text{LL}(\hat{\theta}_A)$: likelihood from ARCH model
  3. Asymptotically χ_p^2 under the null that the models are the same.

name@fit$LLH

```
> # log likelihood for ARCH(1)
> LogL_arch    <- fit_arch1@fit$LLH
> cat("LogLikelihood : ARCH", LogL_arch, "\n")
LogLikelihood : ARCH 10638.59   ARCH.
>
>
> # log likelihood for GARCH(1,1)
> LogL_garch    <- fit_garch11@fit$LLH
> cat("LogLikelihood : GARCH", LogL_garch, "\n")
LogLikelihood : GARCH 11045.76  slide 61
```

Comparing ARCH/GARCH Models Example

- ARCH(1) versus GARCH(1,1)
- IBM daily returns

$H_0: \beta_1 = 0$
$H_A: \beta_1 \neq 0$

- $LL(\hat{\theta}_A) = 10638.59$
- $LL(\hat{\theta}_G) = 11045.76$
- $LR = 2 * (11045.76 - 10638.59) = 814.34$

G - A

- $LR \rightarrow \chi^2_1$ under the null that ARCH(1)=GARCH(1,1); i.e., $\beta_1 = 0$

c.v. $\chi^2_1(0.05) = 3.841$

- Reject this null. for sure.

GARCH (1,1) model fits better than ARCH(1) model.

way way way

name it

```
fit_garch11                         Formal class  uGARCHfit
  ..@ fit  :List of 27
  .. ..$ hessian          : num [1:5, 1:5] 13260104 185 -3088128 -310
  .. ..$ cvar             : num [1:5, 1:5] 7.54e-08 1.72e-09 5.74e-12
  .. ..$ var              : num [1:4529] 0.000591 0.000633 0.0006 0.0
  .. ..$ sigma            : num [1:4529] 0.0243 0.0251 0.0245 0.024 0
  .. ..$ condH            : num 8.64
  .. ..$ z                : num [1:4529] 1.441 0.478 -0.674 2.692 -0.
  .. ..$ LLH              : num 11046
  .. ..$ log.likelihoods: num [1:4529] -1.759 -2.65 -2.564 0.814 -2
  .. ..$ residuals        : num [1:4529] 0.03505 0.01202 -0.0165 0.06
  .. ..$ coef             : Named num [1:5] 5.42e-04 -2.77e-02 5.57e-
  .. .. ..- attr(*, "names")= chr [1:5] "mu" "ar1" "omega" "alpha1"
  .. ..$ robust.cvar      : num [1:5, 1:5] 7.90e-08 -4.09e-07 -9.35e-
  .. ..$ A                : num [1:5, 1:5] 2927.8217 0.0409 -681.8565
  .. ..$ B                : num [1:5, 1:5] 3.08e+03 -4.51 -2.87e+05 -
  .. ..$ scores           : num [1:4529, 1:5] -59.73 -22.02 25.82 -10
  .. .. ..- attr(*, "dimnames")=List of 2
  .. .. .. ..$ : NULL
  .. .. .. ..$ : chr [1:5] "mu" "ar1" "omega" "alpha1" ...
  .. ..$ se.coef          : num [1:5] 2.75e-04 1.55e-02 2.36e-06 7.36
  .. ..$ tval             : Named num [1:5] 1.97 -1.78 2.36 8.86 100.
  .. .. ..- attr(*, "names")= chr [1:5] "mu" "ar1" "omega" "alpha1"
  .. ..$ matcoef          : num [1:5, 1:4] 5.42e-04 -2.77e-02 5.57e-0
  .. .. ..- attr(*, "dimnames")=List of 2
  .. .. .. ..$ : chr [1:5] "mu" "ar1" "omega" "alpha1" ...
```

$\rightarrow LL(\hat{\theta}_G)$